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RESEARCH ARTICLE

Nano-/Micro-fiber Engineering of Vinylene-Linked Polymeric Frameworks for Flexible Free-Standing Thermoelectric Films

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Abstract
Polymer-based thermoelectric (TE) films feature several prominent merits, involving available multi-component compositions, versatile patterning fabrication, and readily integration. Therefore, these materials hold a huge potential as the continuous power supply for wearable devices. Herein, we reported the preparation of a series of vinylene-linked triazole-cored covalent organic frameworks (COFs) by Knoevenagel condensation of 2, 4, 6-trimethyl-1, 3, 5-triazine as the core monomer. The as-prepared COFs tend to generate the nano- or micro-fiber morphologies with tunable lengths and diameters through changing the polyphenylene building blocks. Accordingly, these COF fibers could be readily composited with single-walled carbon nanotubes (SWCNTs) to form the flexible free-standing films upon a simple vacuum filtration method. A film sample containing 30 wt% g-$C_{18}N_3$-COF exhibited the highest power factor of 68.93 μW/(m K^2) at 420 K. The manipulated 4-leg flexible thermoelectric generator (f-TEG) released a maximum output power and power density of 343.5 nW and 0.32 W/m^2 at a temperature difference of 35 K. After bending for 1000 times at a radius of 15 mm, the resistance change rate of the as-fabricated f-TEGs was less than 5%, exhibiting excellent stability and flexibility. This work might not only broaden the potential application scope of COF materials but also provide a new fabrication strategy towards energy harvesting.

Keywords Covalent organic frameworks · Nano-/Micro-fibers · Composite films · Flexible thermoelectric generators

1 Introduction

Flexible thermoelectric generator (f-TEG) can be well-integrated with the wearable and portable technologies, which has the ability to efficiently convert waste heat into valuable electricity [1–3]. Typically, the energy conversion efficiency of f-TEG is dominated by a dimensionless figure of merit (ZT) of the thermoelectric (TE) materials [4], which is given by $ZT = S^2\sigma T/\kappa$, where S, σ, κ, and T are Seebeck coefficient, electrical conductivity, thermal conductivity, and absolute temperature, respectively ($PF = S^2\sigma$ is known as the power factor [5, 6]). The mechanical flexibility of TE materials also plays an important role in the architecture of high-performance f-TEGs [7, 8]. Nanocarbon materials, such as carbon nanotubes (CNTs), possess high electrical properties [9] and outstanding mechanical flexibility [10]. However, CNTs exhibit the disadvantage of poor dispersity during the manufacturing process. Generally, the composition of polymers and CNTs delivered distinct advantages over their individual cases, and various polymer/CNTs TE composites have been studied for the f-TEGs [7]. For example, Huang et al. [11] prepared carbon nanotube film (CNTF)/poly(3,4-ethylene dioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) TE composites, releasing a PF of 339.6 μW/(m K^2) with a corresponding Seebeck coefficient of 67.7 μV/K. The CNTs/polyaniline (PANI) fibers were fabricated through a facile wet-spinning technique. The as-prepared CNTs/PANI fibers showed a PF of 176 μW/(m K^2) and delivered a maximum power density of 7.46×10^{-4} W/m^2 at a temperature difference (ΔT) of 14 K [12]. The carbon nanotube composite yarns (CNTYs) were fabricated with CNTs coating on cotton

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yarns by Jiang et al. [13], which showed a PF of 4.79 μW/(m K^2) after being coated with PEDOT:PSS and delivered a maximum output power density of 280 μW/m^2 at a ΔT of 32 K for the corresponding f-TEG. Recently, covalent organic frameworks (COFs) are emerging as a class of crystalline porous polymer materials [14] and suitable to develop promising TE materials due to their unique characters such as π-extended skeletons, long range-ordered structures, and regular channels, etc. [15]. For example, the fluorene-based covalent organic framework (FL-COF-1) exhibited a high S of 2450 μV/K, and a PF of 0.063 μW/(m K^2) at room temperature (RT) [16]. Deng et al. [17] predicted that the PF of COF poly(tetrathienoanthracene) (PTTA) could be as high as 14.9 μW/(cm K^2) and 21.9 μW/(cm K^2) at 300 K with optimal n-type doping and p-type doping, respectively. Although COFs have broad prospects in f-TEG application, they still suffer from problems such as low electrical conductivity, poor film-formation, and inferior processability, which limit their practical applications [18, 19].

Previously, we successfully developed a new linkage of COFs with carbon–carbon double bond linkages, which exhibited extremely high stability and substantial semiconducting properties [20]. Specifically, combining the triazine units into the frameworks led to the formation of the nanofiber morphologies with the uniform diameters in the levels of nanometers and lengths up to dozens of micrometers [20]. These COF nanofibers could easily form flexible films when mixed with single-walled carbon nanotubes (SWCNTs). The resultant COF-based film composites possess outstanding electrical conductivity and mechanical characters. The COF-based films could be patterned into interdigital electrodes and further assembled into flexible micro-supercapacitors, releasing excellent electrochemical performance and high integrability [21]. However, as far as we know, it is not explored to take advantages of COFs for the development of promising f-TEGs through optimizing their morphologies.

Herein, we report on preparation of a series of vinyl-linked triazine-cored COFs using Knoevenagel condensation and regulation of their nanofibers through the variation of the organic building blocks. Furthermore, the resultant COF fibers were composited with SWCNTs to form the flexible free-standing films upon a simple vacuum filtration. Upon the optimization of the type of COFs and the contents of COFs in the composite films, the TE properties of the composite films were systematically examined and the correlation of the nanofiber morphologies of the as-prepared COFs with their TE properties was rationally established. Finally, the corresponding f-TEGs based on the optimal COF/SWCNTs composite films were manipulated and investigated.

2 Experimental Sections

2.1 Materials

2,4,6-Trimethyl-1,3,5-triazine (TMTA), 1,3,5-tris(4-formylphenyl)triazine (TFPT) and 1,3,5-tris-(4′-formylbiphenyl-4-yl)triazine (TFBT) were purchased from Jilin Chinese Academy of Sciences—Yanshen Technology Co., Ltd. 1,4-diformylbenzene (DFB) was purchased from TCI Shanghai. KOH was purchased from Macklin. The n-butanol (n-BuOH) and 1,2-dichlorobenzene (o-DCB) were purchased from J&K. Anhydrous ethanol was purchased from Adamas-beta Reagent. Acetone was purchased from Shanghai Hushi Chemical Co., Ltd. SWCNTs were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd.

2.2 Synthesis of g-$C_{48}N_6$-COF

In an argon-filled glove box, TMTA (0.25 mmol), TFBT (0.25 mmol), and KOH (0.75 mmol) were dissolved in a binary solvent of 7 mL n-butanol and 3 mL o-DCB and then sealed in a thick-walled pressure tube. This mixture was heated up to 150 °C for 72 h. After cooling to RT, the resulting precipitates were washed with methanol, dichloromethane, tetrahydrofuran, and acetone in sequence for three times (10 mL for each), and then dried under vacuum at 60 °C for 12 h. The sample was denoted as g-$C_{48}N_6$-COF (Fig. S1).

2.3 Synthesis of g-$C_{30}N_6$-COF

In an argon-filled glove box, TMTA (0.25 mmol), TFPT (0.25 mmol), and KOH (0.75 mmol) were dissolved in a mixed solvent containing 7 mL n-butanol and 3 mL o-DCB and sealed in a thick-walled pressure tube. The resulting solution was heated up to 120 °C for 72 h. After cooling to RT, the precipitates were washed with methanol, dichloromethane, tetrahydrofuran, and acetone in sequence for three times (10 mL for each), and then dried under vacuum at 60 °C for 12 h. The sample was denoted as g-$C_{30}N_6$-COF (Fig. S2).

2.4 Synthesis of g-$C_{18}N_3$-COF

In an argon-filled glove box, TMTA (0.50 mmol), DFB (0.75 mmol), and KOH (1.5 mmol) were dissolved in a mixed solvent of 7 mL n-butanol and 3 mL o-DCB. Other synthesis conditions were the same as g-$C_{30}N_6$-COF. The sample was denoted as g-$C_{18}N_3$-COF (Fig. S3).

2.5 Preparation of COF/SWCNTs TE Composite Films

g-$C_{48}N_6$-COF (6 mg) and SWCNTs (14 mg) were separately dispersed in ethanol (20 mL), under ultrasonic treatment for 1 h and 12 h, respectively. The resulting suspensions were mixed together and ultrasonicated for 0.5 h. After that, the mixed suspension was vacuum-filtrated through a membrane filter and the obtained thin film was dried for 12 h at RT to form the free-standing g-$C_{48}N_6$-COF/SWCNTs composite film. The g-$C_{48}N_6$-COF/SWCNTs composite films with weight ratios of 10 mg:10 mg and 14 mg:6 mg were prepared according to the same method. The composite films with mass fractions of 30 wt% (6 mg:14 mg), 50 wt% (10 mg:10 mg), and 70 wt% (14 mg:6 mg) g-$C_{48}N_6$-COF were donated as g-$C_{48}N_6$-COF-30/SWCNTs, g-$C_{48}N_6$-COF-50/SWCNTs, and g-$C_{48}N_6$-COF-70/SWCNTs, respectively. The g-$C_{30}N_6$-COF/SWCNTs and g-$C_{18}N_3$-COF/SWCNTs composite films were prepared as the similar process of g-$C_{48}N_6$-COF/SWCNTs and the corresponding composite films with mass fractions of 30 wt% (6 mg:14 mg) g-$C_{30}N_6$-COF and g-$C_{18}N_3$-COF were donated as g-$C_{30}N_6$-COF-30/SWCNTs and g-$C_{18}N_3$-COF-30/SWCNTs, respectively.

2.6 Assemble of COF/SWCNTs f-TEGs

The optimized g-$C_{18}N_3$-COF-30/SWCNTs composite film was cut into several legs with 5 mm × 20 mm for each, and then stuck on a flexible polyimide film with 5-mm intervals. The four legs were connected by silver electrodes in series and then dried at 60 °C for 6 h to form a g-$C_{18}N_3$-COF-30/SWCNTs f-TEG (Fig. 1).

2.7 Characterization and Measurements

The crystalline phases of samples were tested by an X-ray diffraction analysis (XRD, Bruker D8 Advance, Germany). The micro-morphology was conducted using a scanning electron microscopy (SEM, ZEISS GeminiSEM 300, Germany). Thermogravimetric analysis was detected by a thermogravimetric analyzer (TGA, PerkinElmer STA 6000, American). The morphologies of COFs and SWCNTs were characterized by a transmission electron microscopy (TEM, JEOL JEM-F200, Japan). Fourier transform infrared spectra were recorded with an infrared spectrometer (FT-IR, Perkin Elmer Spectrum 100, American). The S and σ of the composite films were tested by a thin-film TE test system (MRS-3, China) from 300 to 420 K under vacuum with 7% and 5% testing errors for S and σ, respectively. The resistances of the COF/SWCNTs composite films and g-$C_{18}N_3$-COF-30/SWCNTs f-TEG after bending for different cycles were measured by a digital multimeter (VC9807A+). The voltage of the f-TEG was measured by a home-made equipment and corresponding output performance was calculated.

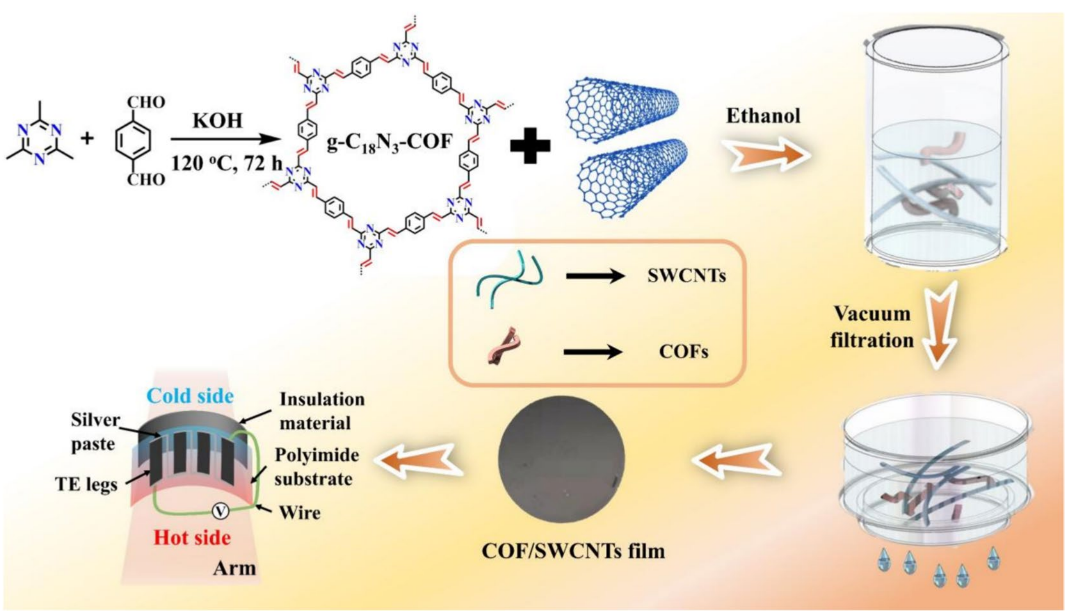

Fig. 1 Schematic illustration of the preparation of g-$C_{18}N_3$-COF/SWCNTs composite films, and its corresponding f-TEGs

3 Results and Discussion

The target COFs termed as g-$C_{48}N_6$-COF, g-$C_{30}N_6$-COF, and g-$C_{18}N_3$-COF were synthesized according to the modified protocols [20, 22] and were previously investigated by ^{13}C NMR spectra [20, 22]. FT-IR spectra analysis revealed that the resonance signals at ~ 1633 and 980 cm^{-1} could be attributed to the trans-vinylene units, indicating the formation of carbon–carbon double bond linkages in the resultant COFs [22] (Fig. S4). Moreover, the XRD pattern of each resulting COFs showed a set of well-resolved reflections (Fig. S5), matched well with the modeling stacking in eclipsed (AA) arrangement [20, 22]. Furthermore, TGA revealed the high thermal stability of g-$C_{18}N_3$-COF powders, with less than 10% weight loss up to 400 °C (Fig. S6).

The microstructures of g-$C_{48}N_6$-COF, g-$C_{30}N_6$-COF, and g-$C_{18}N_3$-COF were presented in SEM and TEM images. The g-$C_{48}N_6$-COF showed rod-shaped hollow structure with the rough surface in the length of several microns (Figs. 2a and S7). The g-$C_{30}N_6$-COF possessed sea cucumber-like morphologies with about 2-μm length and around 200-nm diameter (Figs. 2b and S8). For g-$C_{18}N_3$-COF, the fibers with the several micron lengths and around 200 nm diameter was observed (Figs. 2c and S9). Such phenomenon revealed by microstructural analysis clearly demonstrated that the fiber-like morphologies of these triazine-cored COFs could be tuned with respect to the polyphenylene building blocks in their networks. Our preliminary studies manifested that the presence of the planar electron-deficient triazine units in the 2D skeletons of these COFs could provide strong π–π interactions as the main driving force for the formation of the fibrous assemblies, and the contents of triazine units in the backbones seemed to strongly affect the assembling behaviors, thus varying the fiber-like morphologies [20]. Interestingly, the resulting COF fibers could be suspended in ethanol, which were then mixed with the SWCNTs suspension in ethanol, respectively. Upon a simple vacuum-assistant filtration, a series of free-standing film composites of the as-prepared COFs and SWCNTs can be fabricated. The quality and the thickness of these composite films could be improved and well-tuned by changing the mass amount and content ratios of such two components. Upon the optimization, the thin film with high quality was obtained using COFs and SWCNTs in a ratio of 3:7 w/w, donated as g-$C_{48}N_6$-COF-30/SWCNTs, g-$C_{30}N_6$-COF-30/SWCNTs, and g-$C_{18}N_3$-COF-30/SWCNTs, respectively. The cross-section SEM images shown in Fig. S10 indicated that the average thicknesses of g-$C_{48}N_6$-COF-30/SWCNTs, g-$C_{30}N_6$-COF-30/SWCNTs, and g-$C_{18}N_3$-COF-30/SWCNTs composite films were 58, 55, and 55 μm, respectively.

Figure 2d, g, e, h, f and i presented the TEM images of COF/SWCNTs composites. It can be seen that the g-$C_{48}N_6$-COF, g-$C_{30}N_6$-COF, and g-$C_{18}N_3$-COF were intertwined well with SWCNTs, suggesting the strong interactions between COFs and SWCNTs, and the winding mode of SWCNTs and g-$C_{18}N_3$-COF seemed much stronger (Fig. 2i), which was beneficial to the formation of high-quality composite films. For g-$C_{18}N_3$-COFs, the SEM images of the resulting composite film termed as g-$C_{18}N_3$-COF/SWCNTs showed the good mixed microtexture of the two components (Fig. 2j–l), more or less related to the better match of g-$C_{18}N_3$-COF with SWCNTs either in lengths or diameters. Reasonably, the highly dense and connective textures of g-$C_{18}N_3$-COF/SWCNTs might be favorable for the improvement of TE properties [23, 24]. Obviously, the quality of these composite films is highly dependent on the fiber-like morphologies of the as-prepared COFs.

It is well-known that the morphologies of COFs could significantly affect their processability. Xiong et al. [25] prepared the hollow-structured COFs with high crystallinity and uniform morphologies as well as tunable particle sizes and shell thicknesses by morphological control. They found that the nucleation-growth kinetics of COFs had a significant impact on adjusting their morphologies and structures. The resultant microtube morphologies for such kinds of COFs were primarily driven by the van der Waals interactions between the branched alkyl chains of the molecular building blocks, which were further interwound into the flexible self-standing assemblies [26].

Based on the well-fabrication of the films consisting of COFs and SWCNTs, the electrical conductivities of the COF/SWCNTs composite films with different COF contents were examined. The g-$C_{48}N_6$-COF-30/SWCNTs composite films delivered the highest electrical conductivity and power factor among all the g-$C_{48}N_6$-COF/SWCNTs composite films (Fig. S11). Accordingly, the TE properties of g-$C_{48}N_6$-COF-30/SWCNTs, g-$C_{30}N_6$-COF-30/SWCNTs, and g-$C_{18}N_3$-COF-30/SWCNTs composite films were systematically characterized, as depicted in Fig. 3a–c.

The σ value of 643.94 S/cm at 300 K for g-$C_{18}N_3$-COF/SWCNTs is the highest among the three composite films. When the temperature rose from 300 to 420 K, the S values of the three composite films exhibited an increasing trend (Fig. 3a), whereas their σ decreased gradually (Fig. 3b). Consequently, the PF of the three composite films increased. (Fig. 3c). When the temperature was raised to 420 K, the PF of g-$C_{18}N_3$-COF/SWCNTs obtained a highest value of 68.93 μW/(m K^2). Figure S12 presented the comparisons of PF between g-$C_{18}N_3$-COF/SWCNTs composite films with those in reported references [13, 16, 27–29], the PF of g-$C_{18}N_3$-COF/SWCNTs film was almost the highest value for metal–organic framework (MOF), COF, or their corresponding TE composites. The g-$C_{18}N_3$-COF-30/SWCNTs composite film offered the highest PF value of 68.93 μW/(m K^2) at 420 K among all the composite films, which can

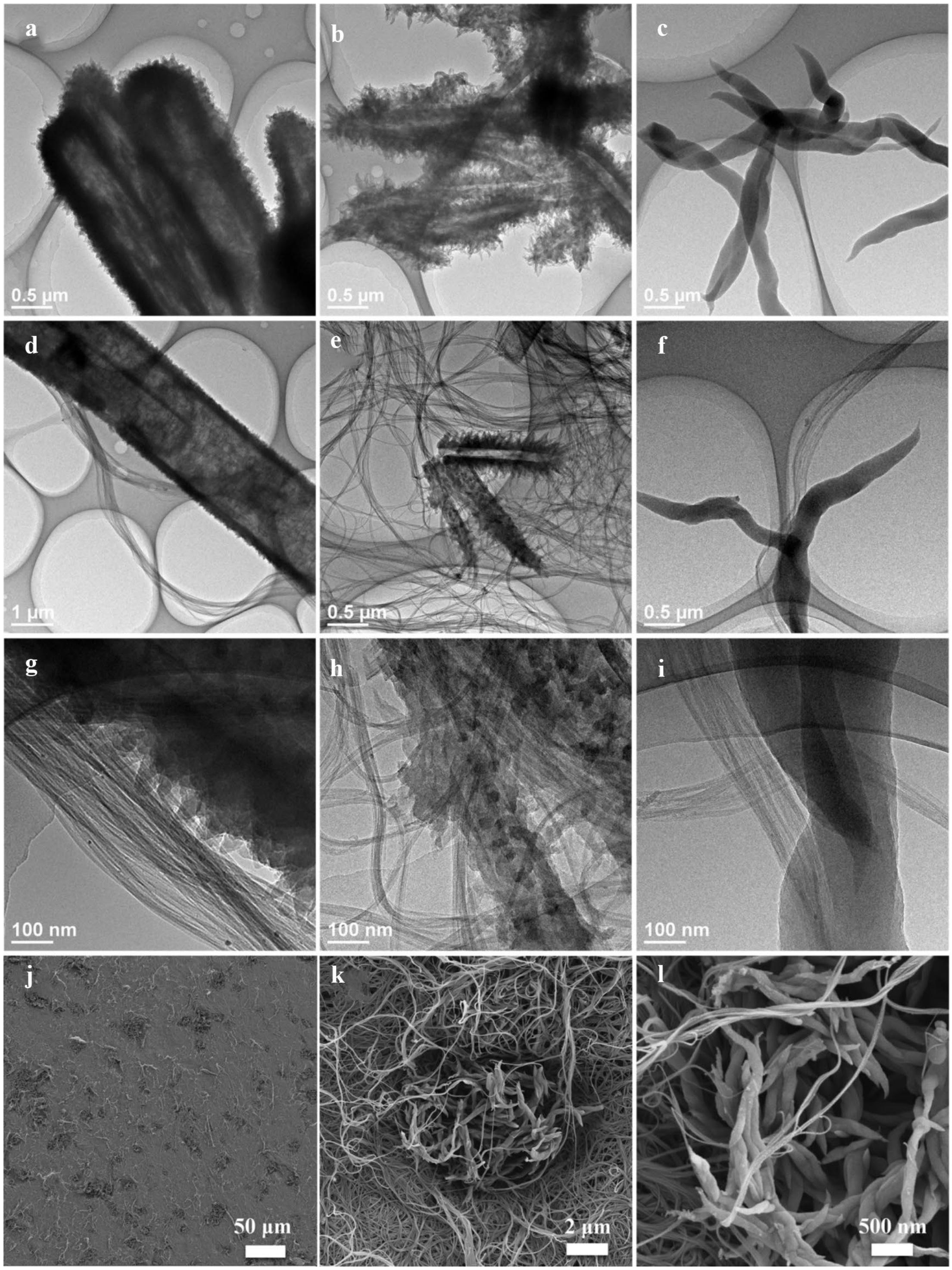

Fig. 2 TEM images of **a** g-$C_{48}N_6$-COF-30, **b** g-$C_{30}N_6$-COF-30, and **c** g-$C_{18}N_3$-COF-30. TEM images of the resultant composite films of **d**, **g** g-$C_{48}N_6$-COF-30/SWCNTs, **e**, **h** g-$C_{30}N_6$-COF-30/SWCNTs, and **f**, **i** g-$C_{18}N_3$-COF-30/SWCNTs. **j**–**l** SEM images of g-$C_{18}N_3$-COF-30/SWCNTs composite films

be ascribed to two reasons as follows: (i) the high σ of SWCNTs is greatly contributed to the σ of the composite film; (ii) Compared with other composite film, g-$C_{18}N_3$-COF possesses more strong interactions with SWCNTs, thus resulting in a much denser texture. The high-dense texture of the as-fabricated films could create much efficient

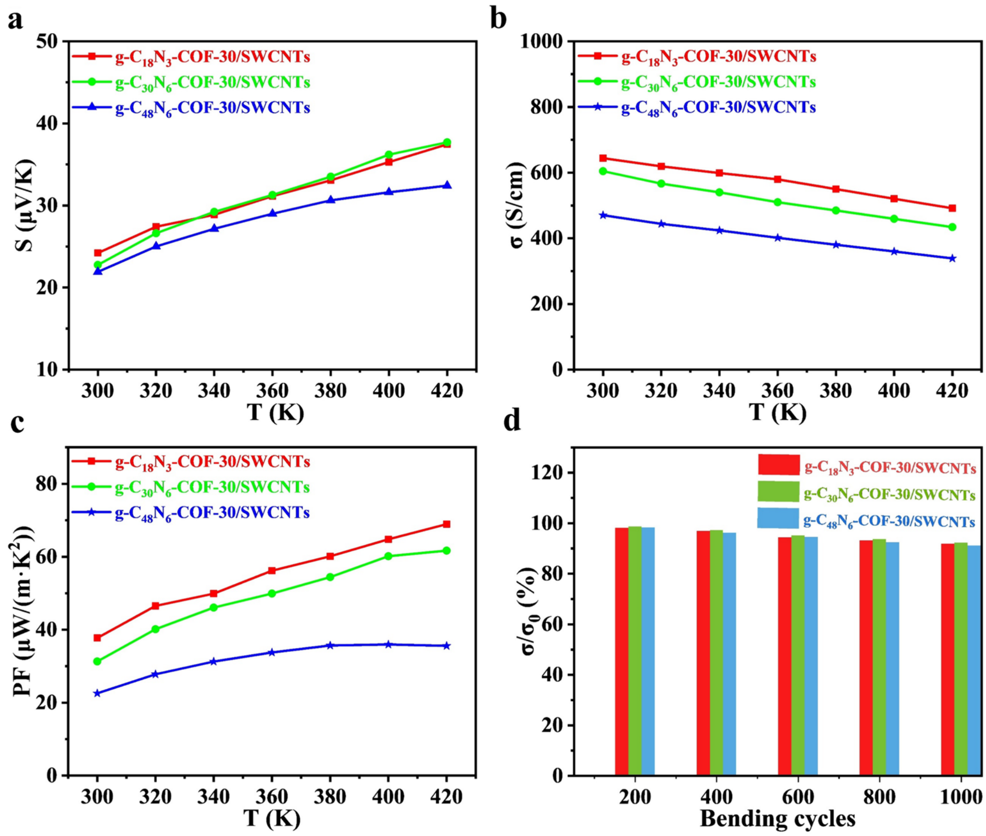

Fig. 3 **a** S, **b** σ, and **c** PF of g-$C_{48}N_6$-COF-30/SWCNTs, g-$C_{30}N_6$-COF-30/SWCNTs, and g-$C_{18}N_3$-COF-30/SWCNTs composite films with different temperature from 300 to 420 K. **d** σ/σ_0 of the composite films as a function of bending cycles at $r=4$ mm

interactions (e.g. π–π interaction) between the COFs and SWCNTs, consequently creating highly conductive pathways. Upon repeatedly bending with a radius (r) of 4 mm for 1000 times, all the resulting composite films still enable maintaining more than 90% of the initial σ values, indicating their good mechanical flexibility (Fig. 3d).

On the basis of their TE behaviors, the application of these COF-based films in f-TEG was further explored. As an example, g-$C_{18}N_3$-COF-30/SWCNTs composite film leg with a dimension of 5 mm × 20 mm for each was used for fabrication of flexible TE devices. The current dependences of the output voltage and output power (P) of g-$C_{18}N_3$-COF-30/SWCNTs-based f-TEG at different ΔT were investigated. The P could be obtained from formula:

$$P = \left[V_{oc}/\left(R_{load} + R_{in}\right)\right]^2 \times R_{load} \quad (1)$$

where V_{oc}, R_{load}, and R_{in} are open-circuit voltage, external load, and internal resistance of a f-TEG, respectively. When R_{load} is equal to R_{in}, the maximum output power (P_{max}) of the f-TEG was obtained as follows [30]:

$$P_{max} = V_{oc}^2/4R_{in} \quad (2)$$

The relationship between the V_{oc} and ΔT shows a good linearity (Fig. 4a). For a 4-leg generator, as $\Delta T = 35$ K, a P_{max} of 343.5 nW (Fig. 4b) and a maximum power density of 0.32 W/m^2 were achieved. The ratio of the change of resistance (ΔR) to initial internal resistance (R_0) as a function of the number of bending cycles of the f-TEG was measured for evaluating the durability of the TE modules. And the $\Delta R/R_0$ value was lower than 5% after bending for 1000 times along the strip length direction at $r = 15$ mm (Fig. 4c), which was mainly due to the good flexibility of the free-standing g-$C_{18}N_3$-COF-30/SWCNTs composite films, indicating good bendable capability of the f-TEGs.

Figure 4d showed the schematic diagram of the output performance test principle and digital photo of flexible g-$C_{18}N_3$-COF-30/SWCNTs f-TEG on the arm. When the flexible g-$C_{18}N_3$-COF-30/SWCNTs f-TEG was attached to the human forearm, a V_{oc} of 0.27 mV was generated. Such f-TEGs assembled by the composite films could convert waste heat into electricity, displaying great potential for powering wearable electronics.

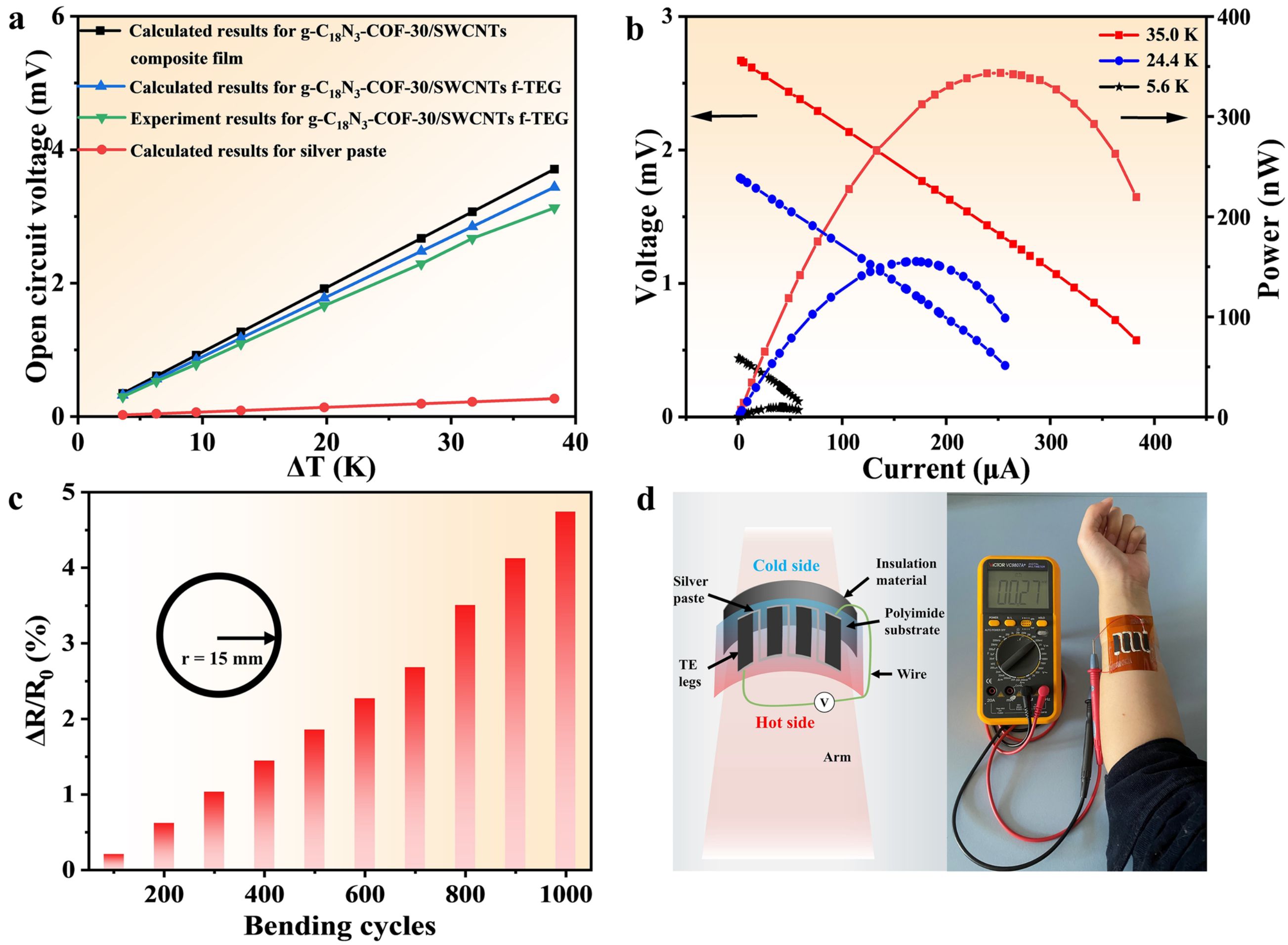

Fig. 4 **a** Experimental and calculated results of the relation between V_{oc} and ΔT. **b** Current dependence of the output voltage and power of g-$C_{18}N_3$-COF-30/SWCNTs f-TEG at different ΔT. **c** The $\Delta R/R_0$ of the f-TEG as a function of bending cycles at $r = 15$ mm. **d** Schematic diagram and digital photo of the output performance for g-$C_{18}N_3$-COF-30/SWCNTs f-TEG

In summary, starting from the preparation of a series of vinylene-linked triazole-cored COFs by Knoevenagel condensation, we have successfully achieved the nano-/micro-fiber morphologies with the tunable diameters and lengths by changing the polyphenylene building blocks in the backbones of these COFs. Moreover, these COFs were composited with SWCNTs to form the flexible films with the tailorable thicknesses via a simple vacuum-assisted filtration. The diameters and lengths of the as-prepared COFs matched with SWCNTs serve as the key role in the high-quality thin films, also favorable for a high thermoelectric performance. A film sample containing 30 wt% g-$C_{18}N_3$-COF exhibited the highest power factor of 68.93 μW/(m K^2) at 420 K, which was further manipulated to a 4-leg flexible thermoelectric generator, releasing a maximum output power of 343.5 nW at a $\Delta T = 35$ K, and a maximum power density of 0.32 W/m^2. The resistance change rate of the as-fabricated flexible thermoelectric generator was less than 5% after bending for 1000 times at a radius of 15 mm, indicating excellent stability and flexibility in good durability of flexible thermoelectric generator. The power factors of the as-prepared COF/SWCNTs composites are almost the highest values for metal–organic framework, COF, or their corresponding TE composites. On the basis of these results, it is reasonable to believe that the efficient interactions between COFs and SWCNTs, not only play a crucial role in the quality of composite films, but also dominate the high performance of the corresponding thermoelectric device. This work for the first time revealed the correlation between the fibrous morphologies of COFs and the properties of the corresponding flexible thermoelectric generator, somehow providing the solid evidence for the development of intriguing thermoelectric films suitable for the integration with modern wearable or portable devices or systems.

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Data Availability Data available on request from the authors.

Declarations

Conflict of interest The authors declare that there are no conflicts of interest.

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